Retaining incomplete river-temperature records: time support of finite-record heat slopes

[Author names][1]

[1][Affiliations]

Key points

- Screening incomplete years changes the time support of sustained-heat facts and of finite-record slopes
- Hidden-peak undercoverage concentrates in Bavarian air reconstructions; extra paths still move those tails without identifying the cause
- Placing years, not merely matching their count, can reduce slope error; restoring all years reduces it further

Abstract

Screening incomplete river-temperature years changes the time support of sustained heat. We examine observed heat windows, finite-record slopes of the annual maximum 30-day mean (H30), and information conditions associated with recovery failures. Whole-archive evaluation uses 42 long complete records from a 581-station source archive. Observed windows in incomplete years can exceed every complete-year heat maximum; most witnesses also pass a warm-season rule. On 805 forward controlled-gap archives, the median complete-record slope is 0.75°C per decade. Recovering all supported years gives slope RMSE 0.170, versus 0.446 under warm-season screening, so the screening error is more than half the median slope. Calendar-position controls that include the fixed complete years reduce RMSE to 0.428 and 0.402; their paired intervals exclude zero. Weak same-count controls do not. Hidden-peak 90% coverage is 67.9% for Bavarian air-ridge reconstructions. Common geometric weighting retains a regional contrast, but support loss and training-history imbalance prevent causal attribution. A trajectory-budget experiment regenerates 2,370 original source-calibration cases and scores the same 27 target station-years. Across three seeds, source-recalibrated median absolute 95th-percentile changes from 20 to 500 nested paths are 0.069–0.117°C in the Swiss subset and 0.061–0.101°C in Bavaria. Five hundred paths do not establish convergence or a preferred estimator. These results support preserving observed heat facts and evaluating year selection, reconstruction and numerical uncertainty for a stated period.

Plain Language Summary

Gaps in a river's temperature record do not necessarily erase a hot year: thirty observed days can already be warmer than every complete year. Most such examples also pass a warm-season screen. In our experiments, the choice of years matters even when the number retained is fixed. Choosing years that better cover the record period reduces slope error, while restoring all supported years improves it further. Recovery failures concentrate in Bavarian air-supported reconstructions, but regional comparisons cannot identify their cause. More simulated paths and recalibration on the original independent source data change uncertainty endpoints and some method comparisons. These findings favor retaining observed heat facts and stating the temporal and information support of any reconstructed summary.

1 Introduction

Stream-temperature records support studies of thermal habitat, climate change and water management (Caissie, 2006; Michel et al., 2020; van Hamel & Brunner, 2024). The annual maximum of a 30-day mean (H30) summarizes sustained heat and is related to the thermal peak used in river-network studies (Beaufort et al., 2022). A monitoring gap can change this statistic through the years it removes, the thermal states it hides, and the information that remains from other years or stations.

Prior work has already shown that partial stream-temperature records can support some annual analyses. Johnson et al. (2021) evaluated missing intervals and imputation for annual mean, phase and amplitude, finding limitations that varied among rivers. Beaufort et al. (2022) used summer observations and climate correction for a multiyear thermal-peak metric. Those results concern different targets: Johnson and coauthors do not supply a universal error baseline for annual H30, and Beaufort and coauthors do not evaluate jointly degraded multiyear archives. The question added here is how completeness screening changes the time support of sustained-heat information, and which observations and river-state information then leave annual recovery and trend propagation without support.

That question has two mechanisms. First, the years that a rule deletes, and the thermal states and time-leverage those years carry, change a slope fitted to the remaining finite record. Second, river thermal response, data processing and source support jointly determine the error of recovering a heat peak inside a gap. The intended use is a comparable finite-record H30 slope on a stated record period, together with fully observed 30-day windows retained as archive facts. If an observed lower bound already exceeds a pre-defined H30 threshold, that exceedance can be kept without a precise annual value. Study thresholds are not ecological-hazard standards, and a 0.5°C reconstruction-error flag is not a calibrated probability at another threshold.

Regression record augmentation and maintenance-of-variance methods have a long hydrological history (Matalas & Jacobs, 1964; Hirsch, 1982; Vogel & Stedinger, 1985). A point prediction, a predictive distribution and a decision to release a value have different targets. Distributional imputation should be assessed for

the downstream quantity rather than only for daily squared error (Näf et al., 2023; Gneiting & Raftery, 2007). We compare a daily plug-in, simple corrections, source-role-calibrated stochastic reconstruction and annual distribution regression, then jointly hide portions of a station's multiyear record before local refitting.

We organize the evidence around three research questions. RQ1: what observed heat information does each completeness rule retain or discard? RQ2: how do the retained year set and annual reconstruction errors combine in a slope for a specified record period? RQ3: under which visible information conditions do hidden-peak recovery errors concentrate? The regional, same-count and trajectory-budget diagnostics were developed after inspecting existing results; they are exploratory rather than preregistered confirmation. The six-station trajectory subset was fixed before that replay, not before the wider study. Its 0.05°C tolerance is a numerical diagnostic, not a hydrological acceptance threshold. Table 1 states the corresponding evidence, denominators and boundaries. Throughout, *calibrated* means an empirical offset estimated on disjoint source-role groups; it does not mean that a target region attains nominal coverage.

2 Data and study design

The source archive contains previously assembled European and US daily observations: 581 physical stations, 185 connected-record groups, 27 regions and 6,170 unique complete station-years. NASA POWER daily air temperature is the common meteorological input (NASA Langley Research Center, 2026). Artificial controlled gaps of 30, 60 or 90 days are crossed with overlap with a training-period climatological peak window; empirical incomplete calendars provide a separate family that is closer to the observed archive. Each of twenty source repetitions redraws masks and group-role composition. In the leave-one-year-out source comparison, fitting, interval calibration, score learning and rule calibration use disjoint connected groups outside each held region. In the original forward source split, HUC17 estimator and NVE interval-calibration roles retain other same-region groups, as detailed below. These repetitions describe variability within the available archive, not twenty independent climates. Figure 1 separates the observation-only, single-year and whole-archive evidence.

Table 1. Three research questions, their evidence and inference limits. Regional, same-count and numerical diagnostics are exploratory, developed after inspecting existing results.

Question	Target	Evidence and denominator	Inference boundary
RQ1: What observed heat information survives each screen?	Reported 30-day window L_y versus a stated accepted record; no reconstruction.	All 2,700 eligible station-years at 96 stations in 34 groups. Rules use a common denominator.	Most witnesses pass the warm-season rule. The 19 accepted windows add facts relative to strict completeness; only 2 excluded windows exceed the warm-season accepted record, by 0.150 and 0.113°C.
RQ2: How do year selection and recovery combine in a finite-record slope?	Total slope error $E = S + R$, with a signed MSE cross term.	807 common forward controlled-gap archives; 805 have actual masked years and enter same-count diagnostics. There are 17 connected groups.	Calendar-CDF and spread controls improve over warm-season selection at fixed counts; all-year recovery improves further. 337/805 selections are fixed by count constraints. Pure count effects and equivalence remain unidentified.
RQ3: Where does hidden-peak recovery fail?	Hidden-peak nominal 90% coverage and recovery bias under visible information conditions.	Regional cases receive common-cell weights with support denominators. A 27-target, two-group numerical subset repeats three seeds and recalibrates on 2,370 original source-role cases.	Bavarian air-ridge failures lack a Swiss counterpart. Training imbalance persists. Re-estimated source offsets change endpoints and local method comparisons; 500 paths do not certify convergence or a general ranking.

We use complete daily station-years as reference outcomes for artificial masking. Water observations hidden by a mask never enter the local fit for that case. Features available when a value is evaluated include the visible gap geometry, a training-period climatological peak, local water history, air conditions and available neighboring stations. The true peak location and hidden water values are evaluation quantities. Observation quality is inherited from the assembled source records; a reported value is not assumed to be error-free physical truth. An observed lower bound is valid for the reported series if its thirty measurements are comparable and valid.

The annual target is

$$H30_y = \max_j \frac{1}{30} \sum_{d=j}^{j+29} Y_{yd}. \tag{1}$$

H30 is used because it summarizes sustained heat. Native seven-day (H7) source refits exist in the repository; they are a different ecological window and are not substituted for H30 here. For an incomplete year, the largest fully observed 30-day mean L_y satisfies $H30_y \geq L_y$ for the same reported series. For 1981–2025 we select

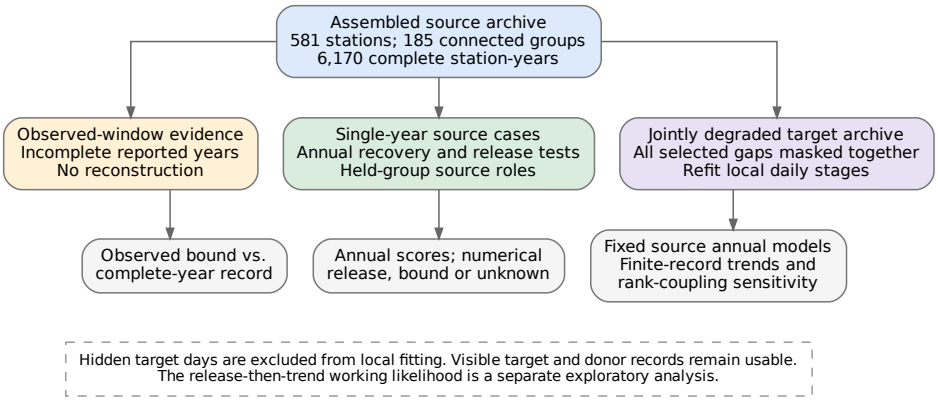


Figure 1. Evidence architecture. Observation-only witnesses, single-year source evaluations and jointly degraded target archives address distinct questions. Hidden target days are excluded from local fitting; unmasked target and donor records remain available. Annual source models are fitted separately and held fixed during each archive reanalysis. Release followed by the working-likelihood trend is a separate exploratory analysis.

stations with at least fifteen complete reference years and compare every incomplete-year bound with the largest complete-year H30. The observation-only denominator contains 2,700 eligible station-years at 96 stations in 34 connected groups. We evaluate whether the same witnesses pass a 95% annual-coverage rule and a 90% station-specific warm-season rule. The five consecutive warm months are obtained from the station's climatology after excluding the evaluated year. These are explicit study comparators; the seven-week missing-period result of Johnson et al. (2021) concerns annual signal parameters, not a validated H30 record rule.

The whole-archive benchmark contains 42 stations with at least twenty complete reference years in seven regions, 18 connected groups and 3,320 station–repetition–setting evaluations (Figure 2; Table 2). One Finnish record ends in 1994, leaving 41 stations for forward evaluation. Twenty-four stations are in Bavaria and ten in Switzerland; that long-complete selection is a monitoring-network filter, not a random river sample. These 42 stations are the long-complete tail of the 581-station source archive: median complete years 31 versus 8 among the remaining 539 stations. Bavaria contributes 24 of 42 evaluation stations against 29 of the remainder; USGS

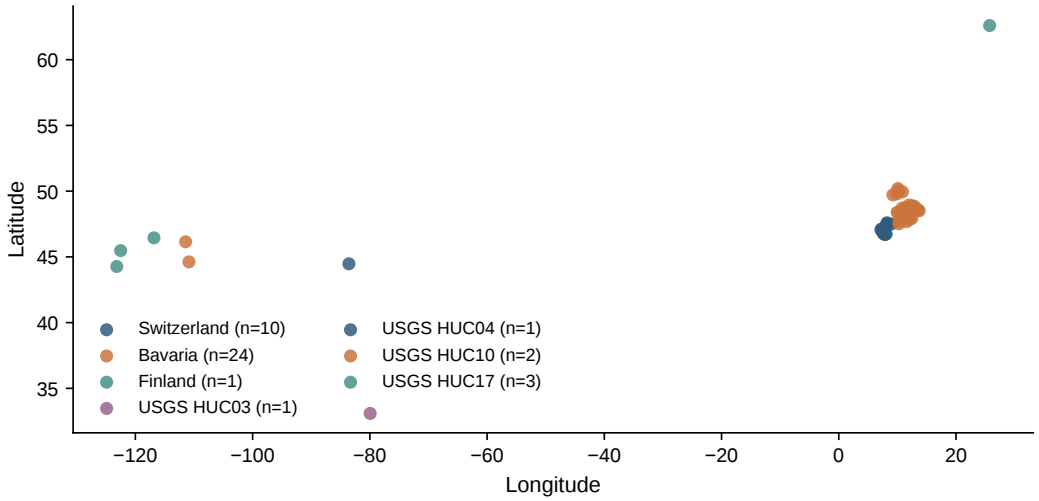


Figure 2. Whole-archive evaluation stations. Long, complete records concentrate in Bavaria and Switzerland; 18 connected groups are analysis clusters, not 18 independent hydroclimates. Coordinates and station codes are in Supporting Information.

regions contribute 7 versus 312 (Table 3). Regulation metadata are unmatched_gages or unknown for most sites and do not mean unregulated rivers; 35 of 42 evaluation stations are unmatched_gages, five are labelled regulated and none unregulated. Daily values are provider daily means where that class is recorded. Visible-neighbour counts and process-metadata gaps are reported with the station roster. If area, elevation or discharge were complete they would be summarized; they are not. For the main retention contrast in the forward controlled-gap setting, both retention sets are available in 807 of 820 settings; the other 820 minus 807 settings are used only for availability accounting. The held-test-group main analysis excludes evaluated test groups from the named source roles. The forward HUC17 estimator and NVE interval calibration retain additional same-region training groups, so the strict-region sensitivity is reported separately. The 2010 split is a retrospective time condition on current-vintage records, not a general nonstationarity test.

3 Methods

3.1 Annual recovery and the decision to release

The daily plug-in computes H30 after a selected daily reconstruction. Functional debiasing corrects its annual value. D2 is a distribution regression of that

Table 2. Whole-archive evaluation stations. Regulation labels are project metadata; unmatched_gages does not mean unregulated. Donor counts are mean visible neighbours in saved leave-one-year-out controlled-gap source cases; the Finnish record is included in that design, not in forward evaluation. Coordinates and station codes are in Supporting Information.

Region	Stations	Groups	Years	Statistic	Regulation	Mean donors
Switzerland (FOEN)	10	2	2000–2025	daily_mean	unmatched_gages	7.3
Bavaria	24	8	1981–2025	daily_mean	unmatched_gages	5.5
Finland (SYKE)	1	1	1935–1994	non_daily_reference_reading	unmatched_gages	0.2
USGS HUC03	1	1	1988–2025	daily_mean	unknown	3.6
USGS HUC04	1	1	1999–2024	daily_mean	unknown	2.9
USGS HUC10	2	2	2001–2025	daily_mean	regulated	3.9
USGS HUC17	3	3	2000–2025	daily_mean	regulated	3.1

Table 3. Long-complete evaluation stations versus the remainder of the 581-station source archive. Complete-year counts are unique complete station-years, including the held-out year, rather than training-year counts. USGS counts follow named hydrologic-unit regions.

Panel	Stations	Groups	Median years	Bavaria	Switzerland	USGS
Trend cohort	42	18	31	24	10	7
Source remainder	539	183	8	29	1	312

annual error: it learns error quantiles from features visible when a value is released, including summaries of simulated trajectories. A stochastic regression comparator generates conditional residual bridges and calculates H30 for each realization. D2 and the stochastic comparator have the same source interval-calibration roles and bootstrap multiplicities. Point estimates are medians; MAE, CRPS, 90% interval coverage and interval score are reported on available cases, with paired comparisons on common cases. Source-role calibration is an empirical offset on disjoint source groups; it is not a certificate that a target region attains nominal coverage.

In the twenty historical source fits, we apply the same observed-window support projection to both methods after their original calibration. The seven stored quantiles constrain only those seven probability levels. Values between them are interpolated, and the outer tails are extrapolated; for comparable CRPS integration we extend the tails using adjacent normal-quantile spacings and enforce L_y when

available. The original model weights and calibration offsets remain fixed in this source re-analysis. In the whole-archive study described below, offsets for both methods are instead re-estimated on their original independent calibration roles after projection. Stored stochastic bridge-repair tables are required to reproduce the historical source, open-source archive and group-maximum sensitivity calculations; these dependencies were checked retrospectively. Neither approach makes the assumed tail shape a verified full distribution.

The conditional-mean mechanism is useful but limited. For observed information X and 30-day averaging vectors w_j, convexity gives

$$\max_j w_j^\top E(Y \mid X) \leq E\{\max_j w_j^\top Y \mid X\}.$$

It concerns the true conditional mean and conditional expectation; an approximate learner need not underestimate every individual year. Calibration, local adaptation, model bias and the chosen point summary remain empirical questions.

We evaluate a rule that reports an estimated value, an observed lower bound or an unknown state. Numerical release requires an available D2, at least four complete historical years, visible features within source-derived limits and a raw individual classifier score no greater than 0.10. The score was trained for a 0.5°C absolute-error flag on the original unprojected D2; that flag is a reconstruction-tolerance diagnostic, not an ecological-hazard threshold, and the score is not a calibrated group-risk certificate for the projected value or for any other threshold. A seven-day close match between visible water and air temperature is tested for the imposed exact-air-substitution scenario; that detector is not evaluated for other stresses. No observed bound is released when this quality flag is raised. The primary factorial trend comparison is separate. A second exploratory analysis passes released values and bounds into a Gaussian working likelihood, treating released numerical estimates as exact and each bound L as a right-censored response with contribution $P(H30 \quad \geq \quad L)$. Unknown cases remain in availability denominators. The resulting regression interval includes working-model variability and is not a task-calibrated missingness-only interval. The three-output rule is retained as a boundary result, not as a proposed general publication system.

3.2 Whole-archive degradation, common cohorts and task-level uncertainty

The whole-archive benchmark jointly masks selected gaps in several years of a target station before fitting its daily models, climatology or residual histories. Unmasked target observations and neighboring physical stations remain available in this experiment. The benchmark contains 42 stations with at least twenty complete reference years in seven regions. One Finnish record ends in 1994, leaving 41 stations for forward evaluation. Twenty paired mask repetitions use both empirical calendars and controlled 30–90-day gaps; after seven observed burn-in years, each reference year is degraded with probability 0.35. These 3,320 archive experiments do not reuse any masked target day when fitting another year. The complete reference, including those hidden days, is used for scoring; its unmasked values can also enter local fitting. The archived single-year plug-in is retained as an optimistic-information paired comparator on identical masks.

Offline daily fits use the remaining visible history. Forward annual models and interval offsets use source observations before 2010 and exclude the evaluated test stations and their connected groups. The original partitioned role assignment can retain other training-only groups in the same named region: this occurs in the HUC17 estimator and in NVE interval calibration. Accordingly this forward experiment is held-test-group, not uniformly held-region, and is applied to 2010 onward. The strict-region sensitivity removes the additional same-region groups and is reported separately. Forward local daily fitting and dependence estimation also exclude water observations from 2010 onward. These are current-vintage retrospective forward experiments, not a historical real-time field trial. Annual regression weights remain source-only and use source-fit seed zero throughout. Source observations do not change across archive masks, so those fits are reused; every local stage that uses the degraded target archive is refitted. Twenty archive masks are not twenty new global model fits.

On each degraded archive, the retention set (90% warm-season rule or all supported years) is crossed with the annual estimator (daily plug-in, equally calibrated stochastic regression or D2). Every slope is scored against the same complete-record slope. For the four MSE cells A (rule/plug-in), B (all/plug-in), C (rule/alternative)

and D (all/alternative), the symmetric retention contribution is $\{(A-B)+(C-D)\}/2$, and the estimator contribution is $\{(A-C)+(B-D)\}/2$. They sum to $A-D$; the interaction is $(A-B)-(C-D)$. Complete paired cells, rather than different available subsets, determine each decomposition. The retention figure uses common station-repetition rows: in the forward controlled-gap setting, both retention sets are available for 807 of 820 settings, and the all-year RMSE is recomputed on those 807 rows.

3.3 Specification of the executed estimators

Daily completion uses one recorded reconstructor per case: donor ridge, air ridge, linear interpolation, smooth climatology, or day-of-year climatology. Air2stream-3 and the value computed from observed days are additional comparators. A reconstructor may use days that remain visible in the target year, the station's other visible years, and eligible neighboring stations. Days hidden by that year's target mask are not used to fit the local model. In the forward experiment, local daily fitting also excludes the target station's water observations from 2010 onward.

D2 learns the annual H30 error relative to the plug-in, not the annual value itself. Seven LightGBM quantile regressions use the quantile objective at levels 0.05, 0.10, 0.25, 0.50, 0.75, 0.90 and 0.95. The fixed settings, which were not re-tuned for this revision, are 300 trees, learning rate 0.05, 31 leaves, at least 50 cases per leaf, 80% row and column subsampling, and an L2 penalty of 1. Features are available at decision time: gap length and runs, warm-season and peak overlap, donor count and correlation, historical reconstruction error, the visible edge anomaly, air anomalies, differences among the other reconstructors relative to the plug-in, and summaries of the stochastic and air-state simulation distributions when those summaries exist. A missing simulation feature stays missing and uses the tree's missing-value branch. Predicted quantiles are sorted to remove crossings. The published annual point is the calibrated 0.50 quantile.

The historical calibrated stochastic comparator stores 20 residual-bridge trajectories as a fixed computational budget. The same-target experiment reported below regenerates the original source-calibration cases at larger budgets and updates their

offsets; it does not rerun the full whole-archive trend comparison. Fine method rankings are not asserted beyond the tested numerical subset. The bridge is a Gaussian AR(1) on a residual history with at least 30 finite days. Its lag coefficient is clipped to $[-0.95, 0.995]$. When an observed water value exists at the left or right end of a gap, the trajectory is conditioned on that boundary; otherwise the left state is drawn from the residual margin. The same seven probability levels are read from those 20 trajectories. Where a stored bridge-repair table applies, it replaces the stochastic columns. It does not refit the daily bridge.

Both annual distributions use four disjoint connected-group roles, with shares 0.55, 0.15, 0.15 and 0.15: estimator fitting, interval calibration, score learning and rule calibration. Groups are shuffled within the source seed and bootstrapped only inside their own role. Each role must contain at least two groups. Interval offsets are empirical quantiles of the calibration-role errors, computed separately by peak-overlap bin and reconstructor. A stratum with fewer than 50 positive-weight calibration cases falls back to all calibration-role cases. Levels at or above 0.50 use the higher empirical quantile, and lower levels use the lower one. Offsets are added and the seven levels are sorted again.

Sampling then follows one order. A finite observed-window lower bound is applied to the annual quantiles. Stratified probabilities are interpolated between the seven stored levels. Outside 0.05 and 0.95, the default extends the quantile function by the adjacent normal-quantile spacing; holding the endpoint fixed is retained only as a paired sensitivity. These seven levels do not identify a complete central 90% distribution: values between them are interpolations, and both outer tails are extrapolations. Samples are then projected onto the same lower bound. Independent ranks, auxiliary Ridge year-bootstrap states, or three-year residual blocks reorder the samples without changing each year's marginal. The ordinary-least-squares slope and its interval are computed from that joint sample. Task-level calibration is a later, separate step.

We refit both methods' annual quantile offsets on their original disjoint source calibration groups after imposing the observed-window lower bound. Both distributions and their samples respect the bound when one is available. Seven stored quantiles constrain only those seven probability levels; they do not identify the

distribution between or outside them. We extrapolate both outer tails using the adjacent normal-quantile spacing and project the annual samples onto the bound; the endpoint-constant alternative is retained on paired cases. The normal extrapolation is an assumption, not a validated tail law. That projection is applied to the annual distribution and then carried into the slope. It does not project a trend interval onto an annual lower bound. The main analysis uses 256 stratified samples per year. Within one estimator, independent ranks, auxiliary shared year-bootstrap regression states and three-year residual blocks keep that estimator's annual marginals fixed. The two estimators do not share one annual distribution. Shared states come from a small harmonic/air Ridge model, not the D2 tree posterior or the selected donor model. These couplings are dependence sensitivities, not posterior samples or coverage theorems.

The joint-rank numerical sensitivity fixes the annual quantiles, observed bounds and evaluation years and increases the stratified sample count through 256, 1,024 and 4,096 on the difficult forward controlled gaps. Three random seeds index the rank construction. The rank sequence is generated at 4,096 draws and truncated for smaller counts; annual marginals are regenerated separately for each sample size. We report coverage, width, interval score, the three-seed mean and the largest single-seed change. A 0.3-point rule on the three-seed mean is only a diagnostic: averaging seeds can cancel opposing changes, and meeting it does not justify ranking small construction differences. The check uses the same opened data and is not an independent validation.

Table 4 separates the calibration and projection order of the historical, whole-archive, task-calibration and external analyses. The task-level comparator calibrates absolute slope error divided by an independent-draw interval half-width. Every source connected group receives equal total weight in the empirical 95th percentile. Calibration uses only unused rule-calibration groups, re-predicted with the held-region model; forward errors end before 2010. We withhold the calibration when fewer than three nondegenerate source groups remain, the calibration factor is effectively zero, or the target half-width lies outside the source 1st–99th percentile range. These thresholds are empirical support checks rather than a finite-sample guarantee. Source support, availability and matched-case width, coverage and interval score

are reported. All uncertainty scores target the finite complete-record slope, not climate-trend uncertainty.

The historical design metadata are not rewritten. The saved 3,320 `design.json` files state that held region and connected groups were excluded; that is correct for held test groups but not an unconditional region exclusion. The forward HUC17 estimator retained one other same-region training group, and NVE interval calibration retained two. The paper uses the held-test-group analysis as primary and the strict-region sensitivity as a separate check. A correction manifest accompanies the historical files.

Two controlled-gap sensitivities assess missingness mechanisms on the first five masks at each of 41 forward-eligible stations. One hides the target's exact gaps from the donor panel; the other preferentially selects air-hot years while preserving the number of missing years. They are retrospective stresses rather than estimates of outage frequency. For the primary gap comparison, all sources use the annual longest contiguous missing run in 2010–2025. The observed archive contains 109 incomplete station-years at 29 of the 41 forward stations, with median 4 days and 11.93% at least 30 days. Empirical-calendar masks have median 6 days and 10.25% at least 30 days over 3806 station-year–repetition rows; controlled masks have median 60 days over 3777 rows, all at least 30 days. Observations receive one weight per station-year; trials receive one per station-year–repetition. These differing samples and repeat weights make this a descriptive comparison. Over 1981–2025 the same statistic has median 3 days, and 11.36% of 220 incomplete station-years last at least 30 days (Table 5). The shared-year measure requires missingness at two connected stations in the same year, not on the same days. Among the 109 observed incomplete station-years after 2010, only 74 have another evaluable cohort station in the same group and year. Of those 74, 38 have a peer that is also incomplete in that year, and 14 have at least one missing day in common with a peer; 40 of 2,086 peer-evaluable missing station-days overlap. The other 35 station-years are excluded from this peer-conditioned denominator. This concerns the 41-station cohort, not every potential or selected model donor. Experimental year selection follows probability 0.35 after a seven-year burn-in; its frequency is not an observed outage estimate. Observation-only accounting uses 96 stations, whereas the whole-archive benchmark uses 42.

3.4 Regional standardization, time-support controls and trajectory budget

The Bavaria–Switzerland comparison reuses saved forward controlled-gap annual cases. Common-support screening retains bins of gap length, training-peak overlap, evaluation year and visible-donor count represented in both regions. It does not balance their frequencies. We therefore standardize the within-cell outcomes to identical weights, proportional to the smaller regional cell count; equal-cell weights give a sensitivity analysis. A second analysis adds reconstructor class and reports its support loss. Total variation of the joint cells, continuous-variable balance within bins and training complete-year balance are reported before and after weighting. Hidden-peak labels use withheld reference H30 and cannot be a deployment filter. No credible flow or regulation subset is available; unmatched_gages is not an unregulated class. Regional and reconstructor associations cannot exclude process explanations.

Complete-year water–air H30 associations are reported both pooled and after centering each variable within station and period (before/from 2010). Paired resampling carries both periods of each connected group; station resampling is a sensitivity. These are annual associations between maxima that need not occur on the same dates, not daily gap-reconstruction coefficients.

Time-support controls reuse saved D2 annual predictions. Complete years remain in every scheme. We first retain the warm-season 90% rule, its original one-draw random subset and its original spacing heuristic for incomplete years. A stronger exploratory comparison enumerates all 5,758 feasible same-count incomplete-year subsets in the 807 original common archives; 5,756 subsets belong to the 805 actually masked archives. Two additional choices use only calendar years and include the fixed complete set in their objective: minimizing the mean squared discrepancy between selected and full-reference empirical year CDFs on the integer-year grid, or maximizing centered year spread S_{xx}. Ties follow the other objective and then lexicographic incomplete-year order. These choices never inspect water temperature or reconstruction errors. Uniform random performance averages squared error over feasible subsets within each archive before taking the across-archive square root. Two hundred fixed random seeds separately describe

selection variation; their percentile range is not a confidence interval. Paired 95% percentile intervals resample the 17 connected groups 10,000 times, retaining every archive in the selected group. The original one-draw contrasts are reported for seeds 20261029 and 2026102969; CDF, S_{xx} and uniform-exact intervals use seed 2026102969. The two seeds agree on whether each legacy interval contains zero. Station- and group-equal weighting are sensitivities. These exploratory intervals are pointwise summaries without multiplicity adjustment across controls and sensitivities. For each selected set, complete-year H30 gives year-set slope error S, annual estimates add reconstruction error R, and total error is $E = S + R$.

The historical classical comparator generates 20 residual-bridge paths. To evaluate budget sensitivity, we freeze the original six stations, 27 scored target station-years, masks, daily means and residual histories, and the original source interval-calibration roles and bootstrap multiplicities. The two target-region roles contribute 1,142 and 1,228 positive-weight calibration cases, 2,370 in total, at 19 stations in 102 station-years before 2010. Each role has four positive-weight connected groups, disjoint from its target group. Native 20-path generation exactly reproduces both the original source repair and target draws and quantiles. Nested 20/100/500 samples are prefixes of a 500-path call for seeds 0, 1 and 2. Because the generator consumes random arrays by day, the native 20-path sample differs from that 500-path prefix; the historical native baseline and nested-budget contrasts are kept distinct.

For every budget and seed we compare three target distributions: no offset, the original fixed source offset, and an offset recomputed on the original source role at that budget. Source quantiles are projected onto the observed bound before weighted calibration, using the original 50-positive-case stratum fallback. Target quantiles are projected, offset, sorted and projected again. No target outcome enters calibration. D2 quantiles remain fixed. All new CRPS comparisons use 256 common stratified probability levels with the same normal-spacing tail continuation, so that scoring approximation is held constant. Coverage, absolute endpoint movements and scores are reported by region/reconstructor; the target sample has only two connected groups. Neither this experiment nor subsequent joint-rank sampling establishes infinite-path convergence or a general method ranking.

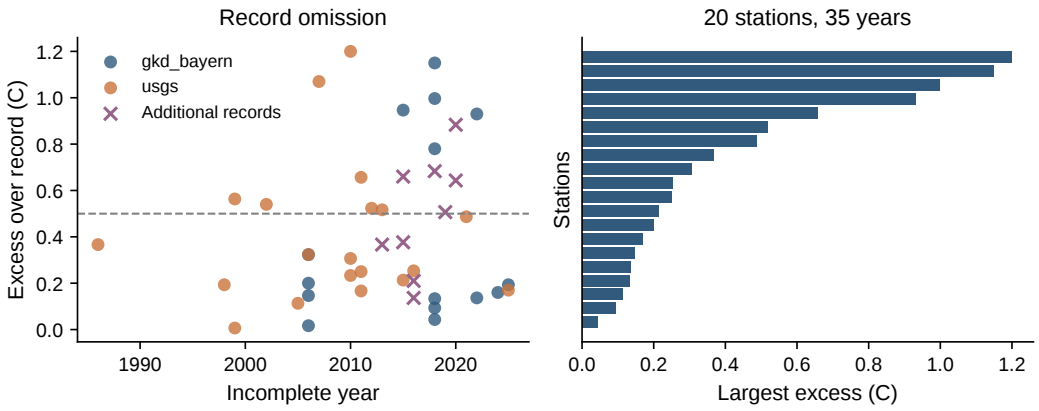


Figure 3. Fully observed thirty-day windows in incomplete years exceed complete-only station records. Crosses on the left are additional approved-record witnesses with sparse one-to-three-year complete reference support; they are not included in the original long-record denominator or the source-station bars on the right. The evidence concerns reported measurements and requires no reconstruction.

4 Results

4.1 Observed windows remain facts under a less restrictive screen

At 20 of 96 eligible stations, 35 incomplete years contain an observed window above the maximum complete-year H30 (Figure 3). Six stations have a witness exceeding the complete-year record by more than 0.5°C. Another incomplete year could still be warmer, and measurement error or a changed sensor location can affect physical interpretation.

The practical consequence depends on the rule. Table 6 accounts for all 2,700 eligible station-years. Complete-year selection admits 2,139 years and discards 561; 545 of the discarded years still contain a fully observed 30-day window and 16 are unknown. The station-specific warm-season rule admits 2,634 years and excludes 66, of which 50 still contain a fully observed window. Of the 35 selected witnesses, 32 pass the warm-season criterion and 31 pass 95% annual coverage. Among the warm-season exclusions, only 2 exceed the largest reported window among the other rule-accepted years, by 0.150 and 0.113 °C. The larger 0.930°C difference for the Bavarian 2022 case is relative to the complete-year record, not to the warm-season accepted record; the two comparisons must not be mixed.

The rule also admits 19 incomplete years whose observed windows exceed the accepted record of the other accepted years. Strict completeness discards all of them. Those 19 windows add information relative to strict complete-year selection. Relative to the warm-season rule, the additional observed facts are the two excluded increments of 0.150 and 0.113°C; remaining incomplete years stay unknown as annual H30. Numerical recovery of those years is a separate cost, reported below. The comparison uses the full archive retrospectively, after setting the evaluated year aside. It is not a sequence of physical records announced in historical time from only the data then available. At an assumed absolute error of 0.05°C on each window, both excluded differences exceed 2ϵ; at 0.10°C, neither difference exceeds 2ϵ. These margins are sensitivity assumptions, not measured instrument errors.

The seasonal-window, past-only and air-regression diagnostics are reported in the Supporting Information. They show that retained heat information depends on the window definition, the historical-support requirement and regulated rivers. They do not reproduce the multiyear climate correction of Beaufort et al. (2022), whose target and site exclusions differ, and they do not establish a universal H30 error baseline from the annual-mean analysis of Johnson et al. (2021).

The additional observation-only USGS intake is reported in the Supporting Information. It corroborates omission at three stations but does not estimate incidence, verify a unique warmest year or establish external reconstruction accuracy.

4.2 Annual recovery is a source-role comparison, not a target-region calibration

On controlled gaps overlapping at least half the training climatological peak, the median source-repetition H30 MAE is approximately 0.428°C for the daily plug-in, 0.355 for functional debiasing, 0.337 for D2 and 0.336 for source-role-calibrated stochastic regression (Figure 4). Median 90% interval coverage is 89.3% and 89.8%, respectively, with CRPS 0.246 and 0.245. These source-calibrated scores describe the executed 20-path classical budget. They establish neither equivalence nor a robust fine method ranking under more fully sampled annual distributions. Both distributions use the same support projection and tail approximation in this comparison; method-specific availability and paired counts are reported in Supporting

Information Table S1. The practical evidence is a gain over the simple plug-in, not a demonstrated universal need for D2.

4.3 Retention, estimator and the common cohort

The whole-archive benchmark comprises 3,320 station/repetition/setting evaluations at 42 long-record stations in 18 connected groups. Figure 5 and Table 7 use common station/repetition rows. For the forward controlled-gap retention contrast, both retention sets are available in 807 of 820 settings; the all-year D2 RMSE on those 807 rows is 0.169 °C per decade, compared with 0.445 under the warm-season rule. The 820-row all-year value is not used for this paired contrast. Retaining years improves the finite-record slope comparison in the evaluated settings. The executed fixed-budget D2 and classical comparisons vary with time setting and region; small differences do not establish a ranking for more fully sampled source-calibrated annual distributions. Results are conditional on donor availability and mask design. Twenty masks at one station are not twenty independent rivers or climates.

The four-cell factorial comparison gives an arithmetic retention contrast. It is not an orthogonal year-set/reconstruction error decomposition or an identified year-count effect. For offline controlled gaps, the symmetric retention contrast is 93.0% of the D2-versus-rule/plug-in MSE gain. This is a result for this experiment, not a universal percentage. Estimator effects and interactions are retained in the full paired table.

The peak-visibility stratification separates cases where the complete reference peak is already visible at the observed lower bound from cases where it is not (Table 8). In the forward empirical-calendar common cases, 3,125 of 3,674 reference peaks are already at the bound and 549 are above it; hidden-peak coverage is 88.3% for D2 and 87.8% for the classical method (Supporting Information). The controlled-gap stratum is harder. In the forward controlled gaps, 1,537 common cases have the reference peak at the bound and 2,146 have a higher reference peak. D2 nominal 90% coverage is 99.3% and 94.4% in the visible stratum, but 83.2% and 81.6% in the hidden stratum. Hidden-stratum MAE is 0.378 and 0.345 °C. Longer gaps, partial training-peak overlap and region change these values: D2 coverage falls from 92.6% at 30-day gaps to 90.3% at 31–60 days and 86.9% at 61–90 days, and from 97.1%

with no training-peak overlap to 83.9% with partial overlap. Bavaria has 82.9% coverage versus 98.7% for the Swiss FOEN stations. Removing Bavaria changes the D2-minus-classical MAE difference from 0.013 to -0.013 °C. These strata are descriptive; the hidden-peak stratum uses the withheld reference and cannot become a deployment filter.

Annual marginals and trend uncertainty give different answers (Figure 6). The left panel is a paired comparison on the two-method intersection, weighted by incomplete years. For forward controlled gaps that intersection has 736 archives and 3,328 missing-year cases; annual coverage is 90.7% for D2 and 88.1% for the classical method. D2 is also available on 818 of 820 archives and the classical method on 736; those availability counts are not the paired denominator. The right panel keeps each method on its own available archives, so the printed sample sizes differ and the panel is not a paired comparison. On D2's own forward set, nominal 95% independent-rank trend coverage is 88.0% and residual blocks give 89.0%. On the intersection, independent-rank trend coverage is 89.1% for D2 and 86.3% for the classical method. The dashed lines are the nominal 90% annual target on the left and the nominal 95% trend target on the right. Within one estimator, the three rank constructions keep that estimator's annual marginals fixed; the two estimators do not share one annual distribution. The joint-rank numerical check increases the samples through 256, 1,024 and 4,096 (Table 9). At 4,096 samples, nominal 95% trend coverage is 87.4%–89.9% across methods, constructions and seeds, so the undercoverage remains. The largest change in the three-seed mean coverage is 0.36 percentage points, and the largest single-seed change is 0.815 percentage points. 5 of 6 method–construction pairs meet a 0.3-point rule on the three-seed mean. That mean can cancel opposing seed changes, so it is not a resolution guarantee and the constructions are not ranked. The experiment does not uniquely attribute the residual undercoverage to annual miscalibration, tail extrapolation or dependence.

Task calibration is available for 39 of 820 forward controlled-gap D2 cases (4.8%), representing two stations and two groups. Both task calibration and the independent-draw comparator cover all of these same cases, but the task-calibrated average width is 0.513 °C per decade versus 0.461 for the independent interval. This selected-subset result does not demonstrate general improvement. Withholding and regional support remain explicit.

The three-output release comparison also does not establish a general safety gain. Matching the numerical output count on future long gaps gives 17,583 values per rule; the three-output rule produces 1,924 errors (10.94%) and simple score ranking produces 1,220 (6.94%). Matching within the 3,174 forward controlled-gap archives gives trend RMSE 0.202 versus 0.196 °C per decade, and the connected-group interval for the paired squared-error difference spans zero. The published-values-plus-bounds working likelihood treats released values as exact and covers 99.8% with a 3.26 °C per decade width; it is not a calibrated missingness-only interval. These are boundary results, not a deployment recommendation.

4.4 Source roles, stress tests and external limits

The main comparison excludes held test stations and connected groups from source roles, but its original forward HUC17 estimator and NVE interval calibration retain other same-region groups. Strict-region and candidate-provider sensitivities are reported in Supporting Information; changing providers, source samples and roles together does not identify a provider effect. These fixed-budget method comparisons do not establish a universal estimator ranking.

Hiding matching target gaps from donors raises D2 finite-record slope RMSE from 0.1555 to 0.2024°C per decade on 205 common forward archives. Matching heat-associated gap templates leaves paired stress effects compatible with zero. These results describe the imposed information conditions and do not identify a physical thermal mechanism or natural outage frequency.

The release rule in Figure 7 withholds unsupported numbers; Figure 8 retains its negative results. It does not beat simple score ranking at matched retention or certify general sensor-quality control. Supporting Information Table S2 reports the source comparison and Table S3 the matched-target comparison; details, the working-likelihood trend analysis and source-role sensitivities remain there.

Strict independent external confirmation remains unresolved. Reference quality, clock, prior exposure and watershed dependence remain separate limits. British Columbia, Nova Scotia, Truckee and reservoir comparisons remain exploratory. Rio Mora's continuous-reference years include 2017 with every quarter-hour slot and bounded years 2012 and 2019; unknown quarter-hour values use the conditional

range $[-5, 50]$ °C. Daily-target accuracy does not establish equivalent physical-reference accuracy. These limits constrain broad transport claims without erasing the observation-only facts or the stated finite-record comparisons.

4.5 Information conditions associated with regional recovery failures

Hidden-peak fractions are similar in Bavaria (58.4%) and Switzerland (56.6%), but their reconstruction support differs (Figure 9; Table 10). Bavarian air-ridge hidden-peak 90% intervals cover 67.9% with bias -0.479°C; 30.2% of references exceed their upper endpoint. Donor-ridge coverage is 90.8% in Bavaria and 98.1% in Switzerland. Daily out-of-fold residual scales are 1.054, 0.673 and 0.455°C, respectively. These are conditional associations among different cases, not a controlled change of reconstructor at the same station.

Common geometric support alone has regional cell-frequency total variation 0.228. Identical min-count weights reduce that joint-cell distance to zero and yield coverage 98.72% in Switzerland versus 82.68% in Bavaria, a 16.04-point difference (Table 11). Equal-cell weighting also retains a contrast. Thus a contrast remains after equalizing the frequencies of these coarse geometric cells. Adding reconstructor retains only 289/1983 Bavarian and 472/1119 Swiss cases. Their standardized coverage difference is 3.69 points, but Switzerland supplies no air-ridge counterpart to the main Bavarian failure layer. Removing 85.4% of Bavarian cases cannot estimate how much reconstructor choice explains the original regional gap. Training complete-year counts also remain strongly imbalanced after weighting (absolute standardized differences exceed 2.7). The cause of the regional difference remains unidentified; process explanations have not been excluded. Deleting one connected group and recomputing common support and weights leaves the geometric coverage contrast positive (12.39–17.77 points), but the reconstructor-conditioned contrast can reverse sign (minimum -1.12 points). This is an influence check over changing supported populations, not a causal contrast.

Pooled annual slopes are 1.143/1.176 in Switzerland and 1.318/1.409 in Bavaria. Within-station slopes are 0.715/0.701 and 0.814/0.809 (Table 12). Pooled correlations rank Switzerland above Bavaria in both periods, 0.842/0.817 versus 0.649/0.708. Within-station correlations keep that order before 2010 (0.855 versus

641 0.826) and reverse it afterward (0.725 versus 0.800; Table 13). The within-station
642 slopes change by -0.014 in Switzerland and -0.005 in Bavaria. The Swiss group-
643 resample interval excludes zero, but two groups produce only three compositions,
644 so that interval is not evidence of a changed water–air relation. Only two Swiss
645 connected groups are available. Neither this limited annual evidence nor maxima
646 on potentially different dates establish stable daily residuals, hidden-peak timing,
647 tail behavior or autocorrelation. In Bavaria, equal station weighting changes the
648 within-slope difference from -0.0054 to -0.0343; deleting one group gives differences
649 from -0.0882 to $+0.1089$. The pooled annual point estimate therefore does not
650 imply uniform station or group behavior.

661 ### 4.6 Year choice matters at fixed counts

662 On 805 actually masked forward controlled-gap archives, the median complete-
663 record H30 slope is 0.754°C per decade (interquartile range 0.496 to 0.906). The
664 warm-season rule retains 9.59 years on average and yields estimated-slope RMSE
665 0.4459, more than half that median. All-year RMSE 0.1697 is less than one quarter
666 of it. The warm-season rule's original one-draw random and incomplete-year-only
667 spacing controls give 0.4540 and 0.4566. On the 468 archives that can select different
668 years, those weak rules give 0.3944, 0.4100 and 0.4150. The spacing rule changes
669 mean span only from 13.265 to 13.343 years, because fixed complete years stay
670 outside its objective. A seed-20261029 group resample gives weak-control minus
671 warm-season intervals [-0.0106, +0.0176] and [-0.0060, +0.0394], both containing
672 zero, and an all-year interval [-0.3572, -0.1832]. Exact uniform averaging over all fea-
673 sible subsets gives 0.4492, with a paired group-resampling interval for its difference
674 from warm-season RMSE spanning zero. This removes single-selection Monte Carlo
675 noise from that comparison. The mean RMSE across 200 separately drawn selec-
676 tions is 0.4505 and their 2.5–97.5% range is 0.4359–0.4671; these describe conditional
677 selection variation, not uncertainty across independent rivers.

678 The stronger calendar controls improve performance (Figure 10; Table 14). Op-
679 timizing the empirical year CDF including fixed complete years gives RMSE 0.4284,
680 with a paired difference interval [-0.0274, -0.0049]; maximizing S_{xx} gives 0.4023 and
681 [-0.0645, -0.0173]. They change 367 and 376 warm-season selections, respectively.
682 Both directions persist with station- or group-equal weights and after each of the 17

group deletions. These exploratory findings show that the earlier weak controls cannot support a blanket claim of no advantage for same-count selection. They assess calendar-based selection rules at fixed count; they neither establish equivalence nor separately identify the effect of adding years.

The count constraints force a single feasible set in 337/805 archives: 332 retain no incomplete year and 5 retain all. The other 468 archives offer a choice. On this subset, warm-season, CDF and S_{xx} RMSEs are 0.3944, 0.3594 and 0.3031. Retaining all supported years remains closer to the specified complete-record slope, with RMSE 0.1697 on all 805 common masked archives, versus 0.4023 for the best tested same-count control. Under seed 2026102969 the paired all-minus-warm-season interval is $[-0.3557, -0.1841]$ °C per decade.

Year-set RMSE under warm-season screening is 0.446, compared with reconstruction RMSE 0.002 on that same set. The all-year truth-set error is zero by definition; its estimated RMSE supplies the nontrivial recovery evidence. The signed identity

$$\mathrm{MSE}(E) = \mathrm{MSE}(S) + \mathrm{MSE}(R) + 2\,\mathrm{mean}(SR)$$

includes error cancellation. For the original random control, $0.206128 = 0.208715 + 0.010617 - 0.013204$. Neither RMSE subtraction nor normalized squared terms identify independent causal contributions.

The original 807-archive paired cohort additionally contains two zero-mask archives, Bavarian stations 10081004/replicate 6 and 11405000/replicate 14, whose empty prediction schemas caused exclusion from the older diagnostic. Restoring their zero errors is a reported sensitivity. The remaining 13 of 820 archives retain only three or four warm-season years and fail the original five-year slope criterion. Their all-year RMSE is 0.3390 and the all-820 all-year RMSE is 0.1735; these unpaired availability results are not substituted for the 805-archive contrast.

4.7 Source recalibration changes the budget comparison

In the original uncalibrated-only nested check, the pooled median absolute upper-end change was 0.0071°C, masking 0.0011°C among 18 Swiss donor-ridge cases and 0.1140°C among nine Bavarian air-ridge cases. 6 of those nine Bavarian upper endpoints, and 7 of 27 overall, move by more than 0.05°C; 10 of 27 do so at ei-

ther endpoint. Mean 90% interval width increases from 0.3179 to 0.3686°C (15.9%). The Bavarian 10046105/2011 upper endpoint moved by 0.4346°C; uncalibrated coverage changed from 23/27 to 25/27. These original diagnostics remain descriptive and reproducible.

The new experiment completes the source-role step. All 2370 native-20 source cases and all 27 native-20 target cases reproduce their historical draws and seven quantiles exactly. Source eligibility remains unchanged across seeds and budgets. Re-estimating source offsets changes the behavior even in the Swiss subset whose uncalibrated median was small (Figure 11; Table 15). Across seeds, the median absolute recalibrated upper-end change from nested 20 to 500 paths is 0.0691–0.1169°C in Switzerland and 0.0609–0.1011°C in Bavaria. The corresponding upper-end exceedance counts are 10–13/18 and 6–7/9. Holding target draws fixed while changing only the source offset separately records the calibration contribution.

On the same 27 targets, historical native-20 classical MAE is 0.1335°C versus 0.1150 for fixed D2. At 500 nested paths with updated source offsets, classical MAE ranges from 0.1100–0.1158 across seeds and crosses the fixed D2 value. Under one common scoring approximation, updated 500-path classical CRPS ranges from 0.0762–0.0787, against historical classical CRPS 0.0879 and fixed D2 CRPS 0.0867. Within seed 0 the updated-source CRPS also crosses fixed D2 between nested 20 and 500 paths. Native 20-path draws and a nested 500-path prefix are different samples. Both historical calibrated methods cover all 27 references, as do the recalibrated 500-path intervals in each seed. These are local comparisons on two connected groups. They show that sampling budget and source recalibration can change an endpoint or a local score. They do not identify a preferred estimator or verify population calibration.

Separately, the original joint-rank experiment increases only samples from fixed annual quantiles and retains nominal 95% trend undercoverage (Table 9). Annual 90% intervals, their 95th-percentile endpoints and trend 95% intervals remain distinct. Supporting Information Table S4 compares the earlier rank and tail assumptions; the new source-budget experiment varies the upstream annual distributions themselves.

5 Discussion

The contribution links observed-window facts, recovery of an annual statistic under stated information conditions, and a slope for a fixed record period. Direct window facts require no imputation. Year selection and annual reconstruction contribute distinct, potentially cancelling errors to the slope. The exact same-count experiment now shows that calendar positions can improve the result even without adding years. Restoring all supported years improves it further in the evaluated archives. This supports attention to time support without assigning a pure causal percentage to year count or treating near point estimates from weak controls as equivalence. The target remains the finite complete-record slope, not an identified climate-process trend.

A regional contrast remains after assigning common weights to the checked geometric cells. History, provider, region and reconstruction support remain entangled, however, and the absence of a Swiss air-ridge counterpart prevents causal attribution. The reconstructor-conditioned contrast can change sign when a group is removed. Similar within-station annual water–air slopes also conceal sensitivity to group and station weighting and do not establish stable gap-day residuals or peak timing.

The completed source-calibration experiment shows why the earlier pooled 0.007°C diagnostic was insufficient. Recalibrated endpoints change in both target strata and local method comparisons depend on the Monte Carlo realization and budget. Historical native20 and a nested500 prefix are different random samples and remain explicitly distinguished. More paths can reduce sampling error without removing structural model or calibration error; 500 paths on two target groups do not prove convergence or transportability. The observation-only facts and D2 year-set comparisons do not depend on declaring a classical-method winner.

The cohort remains the long-complete tail of the archive: median 31 complete years versus 8 among the remaining 539 source stations, with 24 of 42 evaluation stations in Bavaria. Repeated masks and 18 connected groups do not constitute independent hydroclimates. Empirical calendars are closer to observed annual longest-gap conditions than controlled 30–90-day stresses, while actual same-day co-missingness is much lower on the recorded peer-conditioned denominator. That

comparison does not measure every model donor's availability or a common outage cause. External-reference limitations and the negative release and matched-heat-stress results remain in Supporting Information. All added diagnostics are exploratory rather than independent confirmation.

6 Conclusions

Observed 30-day windows in incomplete years can overturn a complete-year heat record. The 19 warm-season-accepted windows add facts relative to strict completeness; two excluded windows add facts relative to the warm-season rule. In the controlled-gap experiment, the median complete-record slope is 0.754°C per decade. Warm-season screening error is more than half of that median, and restoring all supported years brings the error below one quarter of it. Fixed-count rules that place years, including the complete years, can improve on warm-season screening; matching the year count alone does not. Regional standardization retains a geometric contrast without identifying its cause. Repeating both source and target trajectory sampling and recalibrating on the original source roles changes endpoints and local method comparisons, so a small pooled change cannot certify numerical stability. Preserve direct heat facts, specify the record-period target and year-selection rule, and assess reconstruction, source calibration and numerical sampling together.

Open Research

The source observations, processing code, case-level results and figure scripts require a permanent repository and citations to be supplied by the authors. A local review candidate includes processed daily values from 95 of the 96 previously eligible source stations, permitting an independent arithmetic audit of all 35 observation witnesses and their screening-rule decisions. The LUBW station is omitted pending permission, so the full eligible population is not reconstructed. The candidate also includes ten additional USGS daily snapshots and their observation-only replay; neither check reconstructs the main models or verifies subdaily sampling. The Supporting Information lists the access status of each external archive and the derived-file inventory.

The source includes observations from LUBW; its reuse and redistribution terms require provider-specific resolution rather than an assumed open licence.

Although all whole-archive target stations have recorded reuse paths, 13 of 13 source-model estimator roles and 6 interval-calibration roles include positive-weight LUBW cases; the access and sharing conditions of the resulting models and case-level derivatives remain unresolved. Author, affiliation, funding, conflicts of interest, permitted reviewer access and permanent deposit information remain to be supplied before submission.

References

Beaufort, A., Diamond, J. S., Sauquet, E., & Moatar, F. (2022). Spatial extrapolation of stream thermal peaks using heterogeneous time series at a national scale. *Hydrology and Earth System Sciences*, *26*, 3477–3495. doi: 10.5194/hess-26-3477-2022

Caissie, D. (2006). The thermal regime of rivers: a review. *Freshwater Biology*, *51*, 1389–1406. doi: 10.1111/j.1365-2427.2006.01597.x

Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, *102*, 359–378. doi: 10.1198/016214506000001437

Hirsch, R. M. (1982). A comparison of four streamflow record extension techniques. *Water Resources Research*, *18*(4), 1081–1088. doi: 10.1029/WR018i004p01081

Johnson, Z. C., Johnson, B. G., Briggs, M. A., Snyder, C. D., Hitt, N. P., & Devine, W. D. (2021). Heed the data gap: guidelines for using incomplete datasets in annual stream temperature analyses. *Ecological Indicators*, *122*, 107229. doi: 10.1016/j.ecolind.2020.107229

Matalas, N. C., & Jacobs, B. (1964). *A correlation procedure for augmenting hydrologic data* (Tech. Rep. No. 434-E). U.S. Geological Survey. doi: 10.3133/pp434E

Michel, A., Brauchli, T., Lehning, M., Schaefli, B., & Huwald, H. (2020). Stream temperature and discharge evolution in Switzerland over the last 50 years: annual and seasonal behaviour. *Hydrology and Earth System Sciences*, *24*, 115–142. doi: 10.5194/hess-24-115-2020

Näf, J., Spohn, M.-L., Michel, L., & Meinshausen, N. (2023). Imputation scores. *The Annals of Applied Statistics*, *17*, 2452–2472. doi: 10.1214/22-AOAS1727

NASA Langley Research Center. (2026). *POWER project data sets: daily 2 m air tem-*

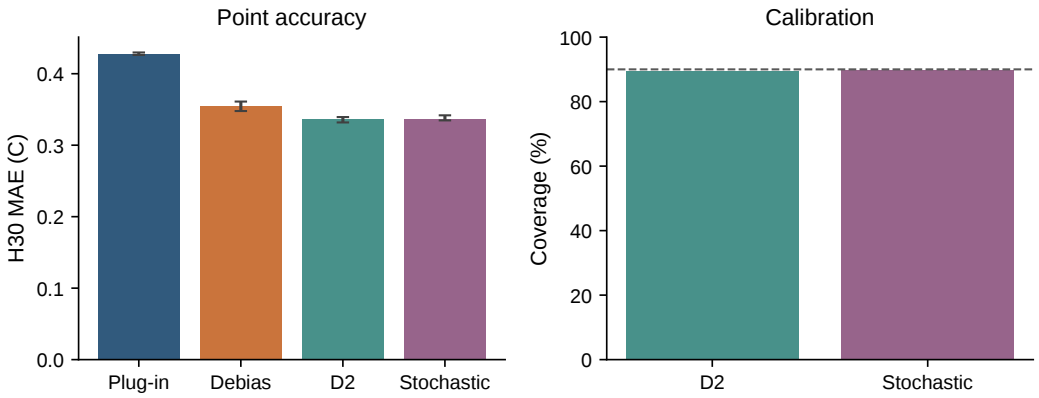


Figure 4. Annual estimation under difficult controlled gaps. Both historical source-calibrated methods receive the same observed-bound projection and normal-spacing tail model. Bars show empirical repetition ranges within the same archive; these are not independent climates.

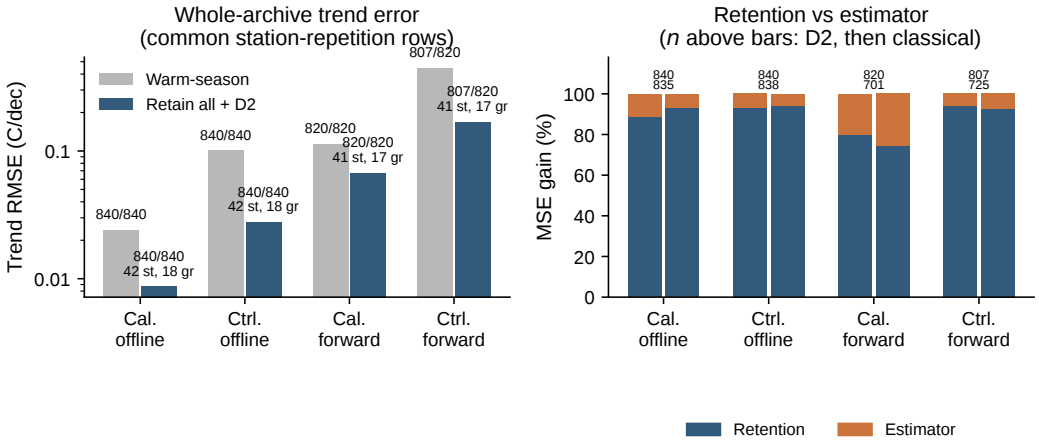


Figure 5. Whole-archive trend errors and the symmetric retention/estimator decomposition. Left: D2 on common station-repetition rows where both retention sets are available; annotations give n and station/group counts. The forward controlled-gap contrast uses 807 of 820 settings. Right: the left bar of each pair is D2 and the right bar is the calibrated classical method. Each bar uses that method's own complete four-cell count, printed under the setting; the two counts are not one shared denominator.

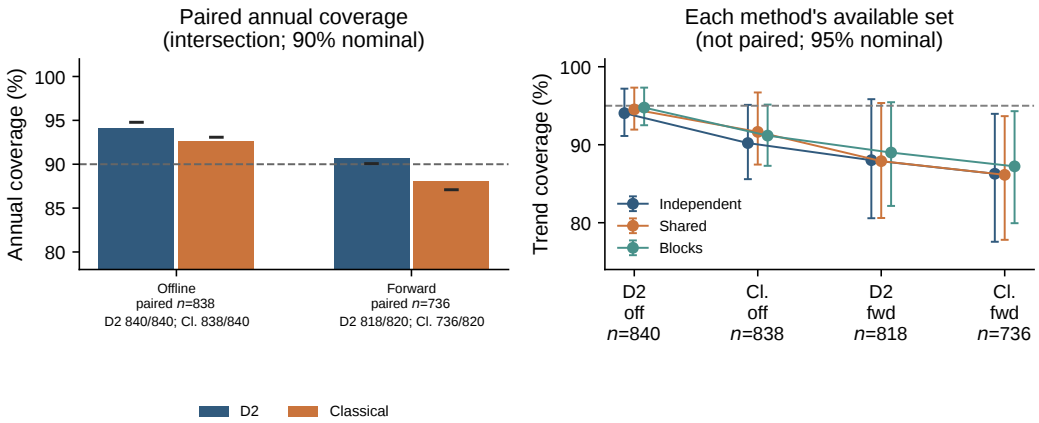


Figure 6. Left: paired annual coverage on the two-method intersection, weighted by incomplete years. Ticks are station-repetition means on the same intersection. The tick labels give the paired archive count and each method's availability out of all archives. The dashed line is the nominal 90 percent annual target. Right: nominal 95 percent trend coverage on each method's own available archives, with that archive count. The right panel is not a paired method comparison. Group-bootstrap intervals use connected groups. No gray band is shown.

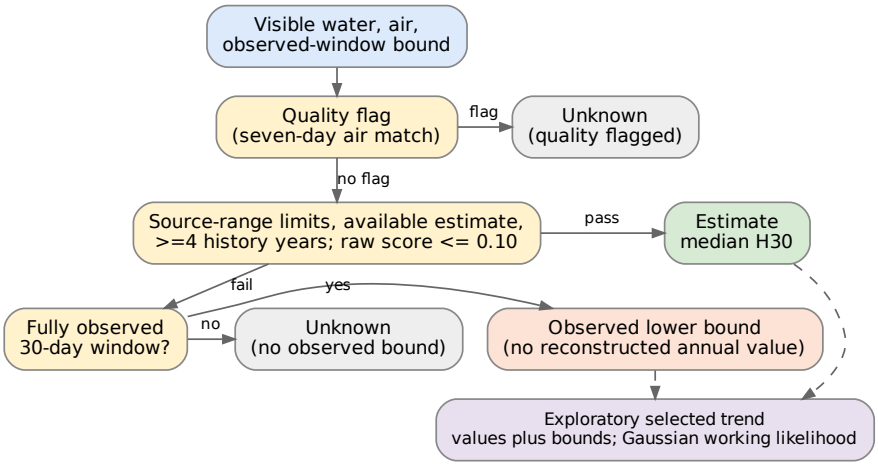


Figure 7. The evaluated three-output decision and exploratory trend analysis. The cap applies to an individual raw classifier score, not a certified group risk. The seven-day water–air check was assessed only for the clean and imposed exact-air-substitution cases. The selected-trend working likelihood treats released values as exact and observed bounds as censored.

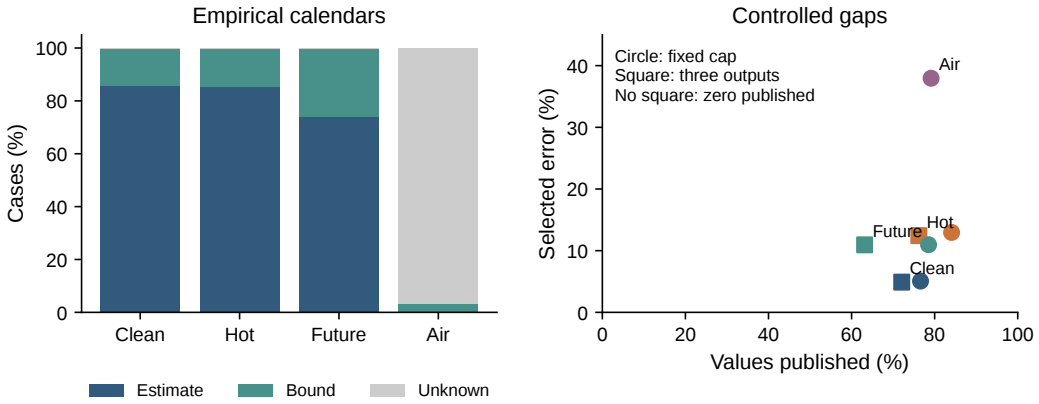


Figure 8. Left: three outputs for empirical-calendar masks. Right: numerical-publication rates and selected errors for controlled gaps. Source diagnostics do not remove extrapolation failures. The air-agreement flag addresses the specified exact-substitution stress only; zero publication is shown as unknown rather than zero-risk success.

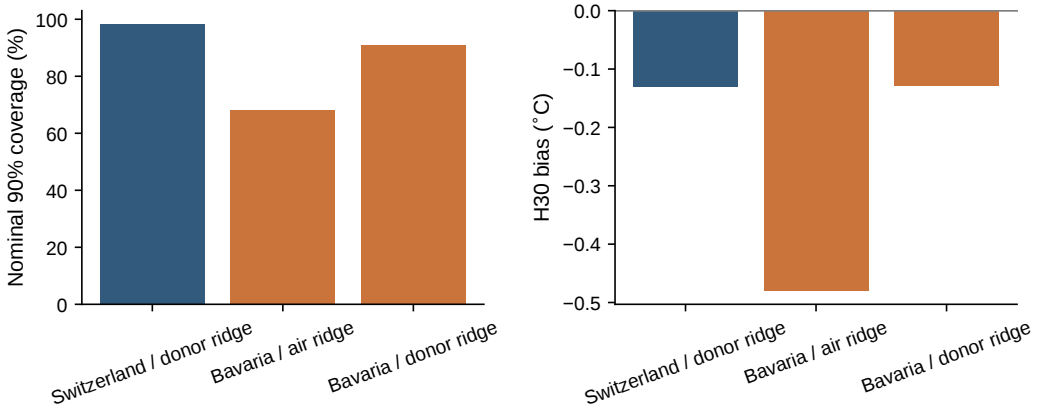


Figure 9. Hidden-reference-peak recovery by region and daily reconstructor. Coverage is nominal 90%; bias is the median minus complete-year H30. The hidden-peak label uses withheld complete-year truth and is unavailable at decision time. Bavarian air-ridge cases are not matched to Swiss donor-ridge cases.

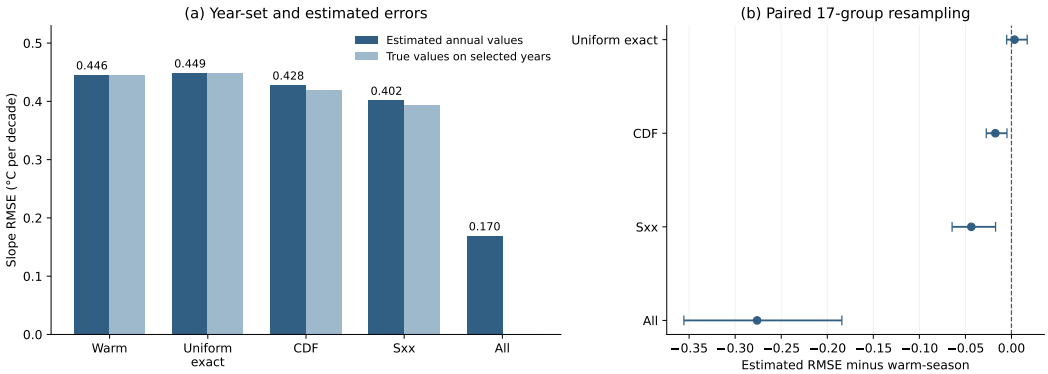


Figure 10. Year choice at a fixed retention count on 805 forward controlled-gap archives. Left: estimated-slope and complete-truth-year-set RMSE. Uniform exact averages squared error over every feasible subset inside each archive before taking the cohort root. CDF and S_{xx} select years without temperature outcomes and include all fixed complete years in their objectives. Right: comparator-minus-warm-season estimated-RMSE differences and paired 95% percentile intervals from 10,000 resamples of 17 connected groups. The original one-draw random and incomplete-year-only spacing controls remain in Table 14.

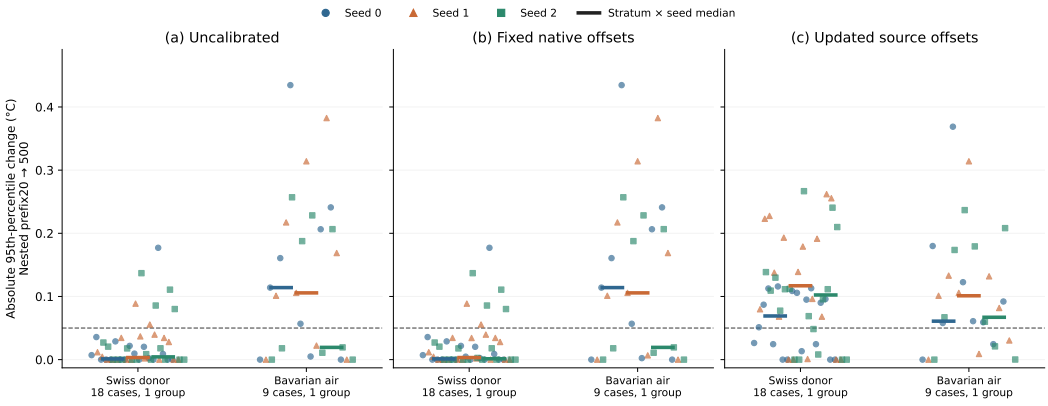


Figure 11. Absolute upper-endpoint changes between nested 20 and 500 residual-bridge paths, for the same target samples under no offset, fixed historical source offsets and source offsets re-estimated at each budget. All 27 case points per seed remain visible, separated into 18 Swiss donor-ridge and nine Bavarian air-ridge cases; colors identify seeds 0, 1 and 2 and horizontal marks their stratum medians. The 0.05°C line is a numerical diagnostic. Source recalibration uses 2,370 original positive-weight cases with unchanged roles and multiplicities. Native historical 20-path draws are a separate reference, not the 500-path prefix. Lower-endpoint changes and all counts are reported in Supporting Information. Two target groups and 500 paths do not establish population calibration or convergence.

perature (MERRA-2). `https://power.larc.nasa.gov/`.

van Hamel, A., & Brunner, M. I. (2024). Trends and drivers of water temperature extremes in mountain rivers. *Water Resources Research*, *60*, e2024WR037518. doi: 10.1029/2024WR037518

Vogel, R. M., & Stedinger, J. R. (1985). Minimum variance streamflow record augmentation procedures. *Water Resources Research*, *21*(5), 715–723. doi: 10.1029/WR021i005p00715

347 **Table 4.** Calibration and projection order in the analyses used here. The rows are separate
348 analyses; they are not twenty repetitions of one final pipeline.

Analysis	Projection and tail treatment	Calibration and source roles	What it supports
Twenty historical single-year source repetitions	The same observed-window projection is applied to both methods after their original calibration. The seven stored quantiles are extended with adjacent normal-quantile spacings.	Original weights and offsets remain fixed. Source roles are the original held-region and held-group assignments.	Historical annual H30 comparisons at the executed 20-path classical budget. Fine differences are not established for larger-budget source-calibrated distributions. Not a whole-archive trend refit.
Whole-archive factorial	The observed-bound projection is applied to each estimator's annual distribution before and after re-estimated offsets, and those annual samples then enter the slope. Each estimator has its own bounded annual marginals. Within one estimator those marginals stay fixed across the three rank constructions. Both use normal-spacing tails. A trend interval is not projected onto an annual lower bound.	Annual weights stay source-only; local daily models, climatology and residual histories are refitted after jointly masking the target archive. The main analysis excludes held test groups; the strict-region sensitivity removes additional same-region groups.	The retention/estimator decomposition and the finite-record trend comparison. The primary retention contrast uses common station-repetition rows.
Forward task calibration	The observed lower bound is imposed on the annual distribution. The independent-draw slope interval is then compared with the task-calibrated slope interval on the same cases. This does not project a trend interval onto an annual lower bound.	Calibration uses unused rule-calibration groups, equal group weights and pre-2010 source errors. Fewer than three nondegenerate groups, a zero factor or an out-of-range target width returns an unsupported status.	The negative result that task calibration is available for only a small selected subset and does not establish population coverage.
External exploratory comparisons	Source models and source calibration are fixed; local history is refitted where the adaptation requires it.	Reference quality, clock, quality codes and watershed dependence are unresolved.	Descriptive transfer evidence only. These rows cannot substitute for a strict independent reference.

Table 5. Annual longest contiguous missing run. The first three rows are the 2010–2025 comparison. Observations weight incomplete station-years equally; trial rows weight incomplete station-year–replicates equally. The 1981–2025 row is the longer observed archive on the same statistic. Sampling and repeat weights differ, so this is not a same-distribution test.

Source	Rows	Stations	Median (d)	90th percentile (d)	≥ 30 d
Observed archive	109	29	4	33.6	11.93%
Empirical calendar	3,806	41	6	30.0	10.25%
Controlled gaps	3,777	41	60	90.0	100.00%
Observed 1981–2025	220	31	3	32.4	11.36%

Table 6. Every eligible station-year under the compared completeness rules. "Number" is a complete-year H30; "bound" is an incomplete year with a fully observed 30-day window; "unknown" has neither. "New facts" counts admitted incomplete years whose observed window exceeds the maximum reported window among other accepted years.

Rule	Eligible	Admitted	Number	Bound	Unknown	New facts	Excluded facts
Complete year (100%)	2,700	2,139	2,139	0	0	0	35
Annual coverage $\geq$95%	2,700	2,591	2,139	452	0	18	3
Station warm-season $\geq$90%	2,700	2,634	2,139	495	0	19	2
Fixed warm-season $\geq$90%	2,700	2,635	2,139	496	0	19	2
July–August complete	2,700	2,509	2,139	370	0	18	3
Longest gap $\leq$49 d (context)	2,700	2,653	2,139	514	0	19	1

A reported window is not a complete-year H30. A rule-accepted incomplete year can contain a higher unobserved peak, so the last two columns compare reported windows only. The 49-day row is a contextual Johnson comparison, not a validated H30 rule.

562 **Table 7.** Whole-archive trend errors on identical available station/repetition rows, in degrees C
563 per decade.

Setting	Reference	Paired n	D2 RMSE	Reference RMSE
calendar / forward	Calibrated stochastic	701	0.062	0.051
calendar / forward	Daily plug-in	820	0.067	0.088
calendar / offline	Calibrated stochastic	835	0.009	0.011
calendar / offline	Daily plug-in	840	0.009	0.014
crossed / forward	Calibrated stochastic	738	0.164	0.147
crossed / forward	Daily plug-in	820	0.173	0.228
crossed / offline	Calibrated stochastic	838	0.028	0.032
crossed / offline	Daily plug-in	840	0.028	0.046

573 **Table 8.** Annual recovery separated by whether the complete reference peak is already visible
574 at the observed lower bound. The stratification uses the withheld complete reference and is
575 unavailable at decision time. Common finite D2 and classical cases, crossed forward controlled
576 gaps.

Reference peak	Method	Cases	Stations	Groups	MAE (°C)	90% coverage	Width (°C)
At observed bound	D2	1,537	41	17	0.012	99.3%	0.17
At observed bound	Classical	1,537	41	17	0.025	94.4%	0.10
Above observed bound	D2	2,146	41	17	0.378	83.2%	1.30
Above observed bound	Classical	2,146	41	17	0.345	81.6%	1.12

577 **Table 9.** Numerical convergence of the three fixed-marginal joint-rank constructions on the
578 difficult forward controlled gaps. Annual quantiles, bounds and evaluation years are fixed. The
579 range is across three random seeds; it is not a confidence interval.

Method	Construction	Draws	Coverage (%)	Width (°C/dec)	Interval score
D2	independent	256	88.3 [88.0, 88.5]	0.450 [0.449, 0.451]	0.855 [0.849, 0.860]
D2	independent	1024	89.2 [89.0, 89.4]	0.457 [0.457, 0.458]	0.845 [0.842, 0.847]
D2	independent	4096	89.2 [89.1, 89.2]	0.459 [0.459, 0.459]	0.842 [0.839, 0.844]
D2	residual blocks	256	89.2 [88.6, 89.6]	0.451 [0.450, 0.451]	0.840 [0.828, 0.851]
D2	residual blocks	1024	89.8 [89.7, 89.9]	0.456 [0.456, 0.456]	0.828 [0.825, 0.830]
D2	residual blocks	4096	89.7 [89.6, 89.9]	0.457 [0.457, 0.458]	0.825 [0.825, 0.826]
D2	shared parameters	256	88.4 [88.3, 88.6]	0.444 [0.444, 0.445]	0.855 [0.845, 0.863]
D2	shared parameters	1024	88.7 [88.3, 88.9]	0.450 [0.449, 0.451]	0.840 [0.835, 0.846]
D2	shared parameters	4096	88.7 [88.5, 88.9]	0.452 [0.451, 0.452]	0.835 [0.834, 0.835]
Classical	independent	256	86.3 [86.0, 86.7]	0.384 [0.383, 0.384]	0.856 [0.849, 0.866]
Classical	independent	1024	87.5 [87.2, 87.6]	0.390 [0.390, 0.390]	0.853 [0.852, 0.853]
Classical	independent	4096	87.6 [87.5, 87.6]	0.391 [0.391, 0.391]	0.851 [0.851, 0.852]
Classical	residual blocks	256	86.9 [86.5, 87.1]	0.384 [0.383, 0.385]	0.840 [0.837, 0.845]
Classical	residual blocks	1024	87.6 [87.4, 87.9]	0.389 [0.389, 0.390]	0.837 [0.835, 0.839]
Classical	residual blocks	4096	88.0 [87.8, 88.2]	0.390 [0.390, 0.391]	0.837 [0.836, 0.839]
Classical	shared parameters	256	86.3 [85.9, 86.5]	0.380 [0.380, 0.381]	0.867 [0.864, 0.868]
Classical	shared parameters	1024	87.3 [87.0, 87.5]	0.385 [0.385, 0.385]	0.856 [0.854, 0.859]
Classical	shared parameters	4096	87.5 [87.4, 87.6]	0.386 [0.386, 0.386]	0.854 [0.852, 0.856]

629 **Table 10.** Hidden-reference-peak recovery by region and daily reconstructor. The hidden-peak
630 label uses withheld complete-year H30. Coverage is nominal 90%. Above-interval is the share of
631 complete references hotter than the 95th percentile.

Region	Reconstructor	Method	Cases	MAE	Bias	90% coverage	Above interval
Switzerland	donor_ridge	D2	633	0.175	-0.130	98.1%	0.9%
Switzerland	donor_ridge	classical	633	0.180	-0.133	96.7%	2.7%
Bavaria	air_ridge	D2	995	0.574	-0.479	67.9%	30.2%
Bavaria	air_ridge	classical	995	0.486	-0.296	66.5%	27.1%
Bavaria	donor_ridge	D2	163	0.249	-0.129	90.8%	8.6%
Bavaria	donor_ridge	classical	163	0.205	-0.079	87.7%	8.6%

Table 11. Descriptive regional coverage standardization on common cells. Min-count weights are the primary common target; equal-cell weights are a sensitivity analysis. Coverage is in percent and differences are percentage points. Geometry cells contain gap, training-peak overlap, year and visible-donor bins. Adding the reconstructor changes the supported population; it does not identify a reconstructor effect.

Cells	Weights	FOEN	Bavaria	Difference
Geometry	Original	98.75	82.78	15.97
Geometry	Min-count	98.72	82.68	16.04
Geometry	Equal-cell	99.29	81.68	17.62
+ reconstructor	Original	99.58	94.12	5.46
+ reconstructor	Min-count	99.72	96.04	3.69
+ reconstructor	Equal-cell	99.75	93.41	6.33

Reconstructor common support retains 472/1,119 FOEN and 289/1,983 Bavaria cases. Switzerland has no air-ridge cases, so Bavaria's air-reconstruction failure stratum has no cross-region counterpart.

651 **Table 12.** Annual water–air H30 associations in unique complete station-years. Years and
652 slopes are shown before/from 2010; S/G counts stations/connected groups. Within estimates
653 center both variables within station and period. Slopes and correlations pool the centered annual
654 observations, rather than averaging station coefficients. The paired 95% percentile intervals
655 resample the same connected groups in both periods; only two FOEN groups and eight Bavaria
656 groups are available, so these intervals are descriptive and coarse. Station-resampling sensitivity
657 results are saved separately.

Region	Fit	Years	S/G	Slopes	Δb [95% interval]
FOEN	Pooled	98/159	10/2	1.143/1.176	+0.033 [-0.005, +0.047]
FOEN	Within	98/159	10/2	0.715/0.701	-0.014 [-0.015, -0.005]
Bavaria	Pooled	554/295	24/8	1.318/1.409	+0.091 [-0.151, +0.346]
Bavaria	Within	554/295	24/8	0.814/0.809	-0.005 [-0.224, +0.149]

The air and water H30 maxima need not refer to the same dates. Similar annual within slopes do not establish stable daily residuals, hidden-peak timing, tails or autocorrelation, and are not an equivalence test.

658 **Table 13.** Annual water–air correlations and paired changes. Intervals use connected-group re-
659 sampling, as in Table 12. Correlations describe annual H30 associations at the stated aggregation
660 scale.

Region	Fit	$r_{<2010}$	$r_{\geq 2010}$	Δr [95% interval]
FOEN	Pooled	0.842	0.817	-0.025 [-0.129, -0.014]
FOEN	Within	0.855	0.725	-0.130 [-0.130, -0.129]
Bavaria	Pooled	0.649	0.708	+0.059 [-0.027, +0.100]
Bavaria	Within	0.826	0.800	-0.025 [-0.066, +0.051]

Table 14. Exploratory finite-record slope comparisons on 805 actual-mask archives in 17 connected groups. Errors are °C per decade. CDF and Sxx optimize the entire retained set including fixed complete years. Uniform exact averages squared errors over feasible same-count sets within each archive, then averages archives and takes the square root. All-year truth-set error is zero by definition. The last column gives comparator-minus-warm-season estimated RMSE with a paired 95% group-percentile interval, not an equivalence test.

Rule	Years	Estimate	Year-set	Reconst.	Δ RMSE [95% interval]
Warm-season	9.59	0.4459	0.4458	0.0023	Reference
Legacy random	9.59	0.4540	0.4569	0.1030	0.0081 [-0.0103, 0.0177]
Legacy spacing	9.59	0.4566	0.4539	0.0873	0.0108 [-0.0063, 0.0404]
Uniform exact	9.59	0.4492	0.4483	0.0978	0.0033 [-0.0052, 0.0171]
CDF	9.59	0.4284	0.4199	0.1079	-0.0175 [-0.0274, -0.0049]
Sxx	9.59	0.4023	0.3931	0.1185	-0.0436 [-0.0645, -0.0173]
All years	13.32	0.1697	0.0000	0.1697	-0.2762 [-0.3557, -0.1841]

740 **Table 15.** Same-target annual comparison by region–reconstructor stratum. D2 and historical
741 native20 are fixed original references. Updated rows regenerate the source-role offsets at the
742 stated nested budget; entries are minima–maxima across seeds 0, 1 and 2, not confidence inter-
743 vals. MAE and mean 90% width are in °C. CRPS uses the same seven-quantile normal-spacing
744 256-point approximation for every method. Each stratum contains one connected group; these
745 counts do not estimate population coverage or establish a general method ranking.

Method / paths	Covered	MAE	Width	CRPS256
Switzerland / donor ridge; 18 target station-years				
Fixed D2	18/18	0.0569	0.7347	0.0483
Historical native20	18/18	0.0785	0.5623	0.0554
Updated nested20	18 of 18	0.0632–0.0666	0.5802–0.6716	0.0504–0.0509
Updated nested100	18 of 18	0.0642–0.0724	0.5677–0.5715	0.0490–0.0521
Updated nested500	18 of 18	0.0693–0.0730	0.5334–0.5533	0.0500–0.0516
Bavaria / air ridge; 9 target station-years				
Fixed D2	9/9	0.2311	1.0531	0.1634
Historical native20	9/9	0.2435	0.7612	0.1530
Updated nested20	8–9 of 9	0.1914–0.2321	0.7357–0.8804	0.1281–0.1621
Updated nested100	8–9 of 9	0.1824–0.2129	0.7526–0.8547	0.1262–0.1467
Updated nested500	9 of 9	0.1915–0.2015	0.7607–0.8006	0.1287–0.1334